

SOFTWARE ENGINEERING

Lab 1: Requirements Engineering & UML Use-Case Modelling

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SECTION: 5A

REQUIREMENTS TABLE:

Problem Statement #53 - Freelance Content Creator Escrow Platform (Media, Events & Community)

Actors: Content Creator, Client Sponsor

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow a Content Creator to submit a draft media deliverable (image/video) for a defined milestone, which the system automatically routes to the assigned Client Sponsor for review.	High	Pass: Uploaded draft appears in Sponsor's review queue within 10 s with correct milestone tag. Fail: Draft not linked to milestone or not visible to Sponsor.	Given sample FR-001; core deliverable-submission workflow of the platform.	Specify max file size/format supported (e.g., MP4, PNG, JPG) and behaviour if upload fails midway.
FR-002	Functional	The system shall allow a Client Sponsor to approve or reject a submitted draft deliverable with an optional comment before any milestone payment is released.	High	Pass: Sponsor decision (Approve/Reject) is recorded against the milestone and status updates to 'Approved' or 'Revision Requested'. Fail: Milestone status changes without a recorded Sponsor decision.	Enforces the sign-off gate required before funds move, per platform's escrow model.	Clarify whether the comment is mandatory on rejection and if there's a time limit for the Sponsor to respond.
FR-003	Functional	The system shall automatically apply a visible watermark to all draft media assets at the time of upload, before the file is made accessible to the Client Sponsor.	High	Pass: Every draft file opened by the Sponsor shows the watermark overlay; original unwatermarked file is not downloadable pre-approval. Fail: Any draft is viewable/downloadable without a watermark.	Protects the Creator's unpaid work from misuse prior to milestone approval.	Define watermark placement/opacity standard and confirm it applies to both image and video previews.
FR-004	Functional	The system shall release the escrowed milestone payment to the Content Creator's balance automatically within 60 seconds of the Client Sponsor's approval sign-off.	High	Pass: Creator balance reflects the milestone amount within 60 s of approval, with a transaction log entry. Fail: Funds released without approval, or delayed beyond 60 s.	Directly derived from sample FR-001; guarantees timely, conditional payment release.	Need to specify which payment gateway/provider is used and the timeout/retry behaviour if it doesn't respond.
FR-005	Functional	The system shall allow a Client Sponsor to request a revision on a rejected draft, specifying feedback text, and notify the Content Creator so a new draft can be resubmitted against the same milestone.	Medium	Pass: Rejected milestone moves to 'Revision Requested' state with feedback text visible to Creator, and accepts a resubmitted draft. Fail: Creator cannot resubmit or feedback is not delivered.	Supports the realistic iterative review cycle between Creator and Sponsor.	Consider capping the number of revision cycles allowed per milestone to avoid disputes dragging on indefinitely.
NFR-001	Non functional	Uploaded media drafts must complete automated watermarking in under 5 seconds for video/image previews, even under simulated peak load.	High	Pass: Benchmark tests confirm watermarking latency < 5 s and security standards hold under simulated peak load. Fail: Any sampled watermarking operation exceeds 5 s or bypasses security checks.	Given sample NFR-001 (Performance & Security); keeps the review workflow responsive.	Define 'peak load' numerically (e.g., concurrent uploads/sec) so the benchmark is repeatable.
NFR-002	Non functional	The system shall encrypt all escrowed funds data and milestone payment transactions using industry-standard encryption (e.g., TLS 1.2+/AES-256), both in transit and at rest.	High	Pass: Security audit/log confirms all payment and escrow records are encrypted in transit and at rest with no plaintext exposure. Fail: Any escrow/payment record found unencrypted in transit or storage.	Financial escrow data requires strong confidentiality and integrity guarantees to maintain user trust.	Name the specific compliance standard to target (e.g., PCI-DSS) since 'industry-standard' alone isn't testable.